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REVIEW



The impacts of menopausal hormone therapy on longer-term health consequences of ovarian hormone deficiency

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ABSTRACT

This study on the longer-term health consequences of ovarian hormone deficiency (OHD) received the Henry Burger Prize in 2022. Osteoporosis, cardiovascular disease and dementia are major degenerative diseases that are also causally associated with OHD. Two randomized controlled trials (RCTs) revealed no significant difference in bone mineral density by adding alendronate to ongoing menopausal hormone therapy (MHT) or combining alendronate at MHT initiation. Another RCT pursuing the effects on fracture recurrence and total mortality in women with hip fracture disclosed that MHT with percutaneous estradiol gel (PEG) and micronized progesterone (MP4) was comparable to risendronate. Basic studies reported that 17 β -estradiol exerted direct beneficial actions on vascular smooth muscle in cell proliferation, fibrinolysis and apoptosis. A fourth RCT showed that MP4 had a neutral impact on the PEG response of blood pressure and arterial stiffness. A fifth RCT suggested that the combination therapy of conjugated equine estrogen and MP4 was superior to tacrine in preserving activities in daily living in women with Alzheimer's disease. In addition, PEG plus MP4 attenuated cognitive decline in women with mild cognitive impairment in a sixth RCT. Finally, the all-cause mortality in recently menopausal women receiving MHT was updated using an adaptive meta-analysis of four RCTs.

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Menopausal hormone therapy; osteoporosis; hip fracture; cardiovascular disease; arterial stiffness; Alzheimer's disease; mild cognitive impairment; estrogen; vascular smooth muscle

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Introduction

Degenerative diseases are a growing matter of public concern in an aging society. Osteoporosis [1], cardiovascular disease such as coronary heart disease (CHD) [2] and stroke [3,4], and dementia [5] are common degenerative diseases in women that are reported to be closely linked to a decline in estrogen levels resulting from menopause, which have been termed 'late problems' of ovarian hormone deficiency (OHD).

The Women's Health Initiative (WHI) study (mean participant age 63 years) demonstrated that menopausal hormone therapy (MHT) employing conjugated equine estrogen (CEE) and medroxyprogesterone acetate (MPA) prevents bone mineral density (BMD) loss and fracture in healthy postmenopausal women [6,7]. However, a significantly increasing time trend in the risk of CHD was found with the number of years since menopause at MHT initiation [8], and the stroke risk was elevated irrespective of the age at recruitment [9,10]. A WHI memory study additionally showed an increase in the dementia risk in older women [11]. Of note, a meta-analysis of prior randomized controlled trials (RCTs) reported that MHT using oral estrogen significantly decreased the CHD risk by 48%, but did not influence the stroke risk in women less than 10 years after menopause [12]. Prospective studies also reported that when MHT is started early after menopause, it

might reduce the risk of dementia [13]. Accordingly, a timing hypothesis has been raised for CHD and dementia. Even though OHD is causally associated, current data reveal a limitation of MHT's efficacy in the prevention of late problems of OHD.

MHT was originally developed for the treatment of climacteric problems, such as vasomotor symptoms and genitourinary syndrome of menopause, which are often called 'earlier problems' of OHD. Currently, the lowest effective estrogen dose for the shortest duration is recommended for the treatment of menopausal symptoms with MHT due to concerns about potential adverse effects of breast cancer as well as cardiovascular disease and dementia. Newer regimens comprising non-oral estrogen and natural progesterone are expected to improve the efficacy and safety of MHT [14].

This study received the Henry Burger Prize in 2022. This report will briefly review the research results of our group on the impacts of MHT on the late problems associated with OHD.

Osteoporosis and fracture

In cases of severe osteoporosis or a suboptimal response to osteoporosis treatment, combination therapy using

medications with different pharmacologic actions might be of benefit. A Finnish study reported that up to 26% of women might experience a densitometric non-response to MHT [15]. A beneficial effect on spine BMD was reported by adding bisphosphonate to ongoing MHT [16]. However, it is still controversial whether the combination of MHT and bisphosphonate could yield a better BMD response than MHT alone [17].

We carried out a placebo-controlled RCT to investigate the effect of added alendronate for 1 year to ongoing MHT in postmenopausal women ($n = 139$, mean age 60 years) with a low BMD (T -score < -2.0) [18]. A better BMD response to the added alendronate was not reproduced in Korean women. Another multicenter, open-label RCT was conducted to examine the effect of combination therapy of MHT and alendronate on BMD in postmenopausal women ($n = 344$, mean age 59 years) with low BMD [17]. One year of combination therapy was not superior to MHT alone in Korean women.

According to the WHI study, MHT reduces the fracture risk in postmenopausal women regardless of age, fracture risk assessment tool probability or fall history [6,7,19]. Of importance, the proportion of WHI participants with osteoporosis of the hip was around 5%. Unfortunately, the anti-fracture efficacy of MHT remains unestablished to date in women with osteoporosis.

We designed an open-label RCT to assess the effects of 4-year MHT on the fracture recurrence rate and total mortality in postmenopausal women ($n = 223$, mean age 75 years) with recent hip fracture in comparison to oral risedronate (35 mg) taken weekly [20]. The MHT group received percutaneous estradiol gel (PEG; 0.1% 1.5 g) plus oral micronized progesterone (MP4; 100 mg) daily. Kaplan–Meier analysis showed no significant differences in fracture recurrence (Figure 1(A)) and

death (Figure 1(B)) between the two groups. Total hip BMD did not change in the risedronate group, but increased significantly by 2.8% in the MHT group (Figure 1(C)).

Cardiovascular disease

Even though the underlying mechanisms are not fully understood, estrogen exerts cardioprotective action via direct arterial impacts that are greater than its systemic beneficial effects on major risk factors, including dyslipidemia, hypertension and diabetes mellitus [21]. We further investigated the direct effects of estrogen on cultured vascular smooth muscle cells (VSMCs) in a rat model. The VSMCs were treated with lysophosphatidylcholine (lysoPC), an active component of oxidized low-density lipoprotein.

The abnormal proliferation of VSMCs is one of the hallmarks of atherosclerosis and coronary restenosis. Cell proliferation was assessed by the incorporation of [3 H]-thymidine. The results indicated that 17β -estradiol (17β -E2) suppressed the DNA synthesis induced by lysoPC via a non-genomic antioxidant mechanism [22].

Thrombosis plays a crucial role in the pathophysiology of atherosclerosis. We reported that 17β -E2 reduced the expression of plasminogen activator inhibitor type 1 (PAI-1) stimulated by lysoPC through a non-receptor-mediated mechanism via antioxidant activity as well as a receptor-mediated mechanism. Further, 17β -E2 suppressed PAI-1 activity and concurrently enhanced tissue-type PA activity as assessed by fibrin overlay zymography, suggesting a favorable action on fibrinolysis [23].

Apoptosis of VSMCs is strongly implicated across all stages of atherosclerosis and is associated with arterial aging. Apoptosis was induced by lysoPC and was assessed using propidium iodide staining; 17β -E2 reduced apoptotic cell

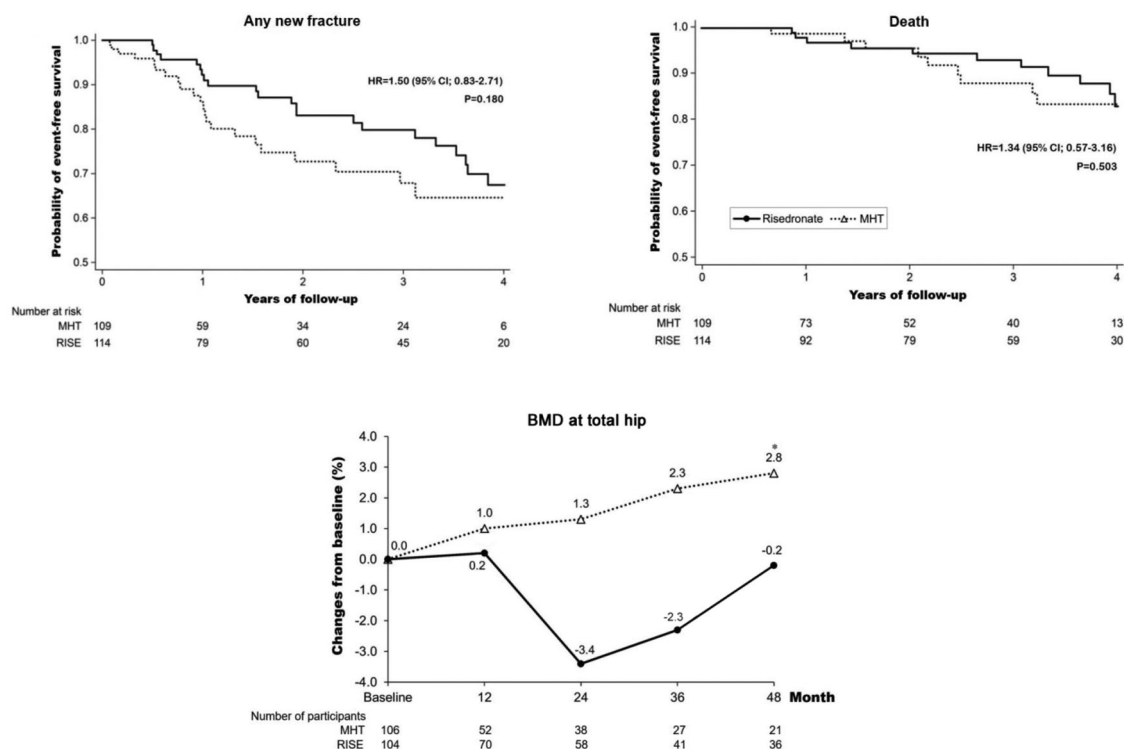


Figure 1. Kaplan–Meier survival curves of any new fracture (A) or death for any reason (B) and the mean percentage changes in bone mineral density (BMD) at the total hip (C). Hazard ratios were assessed among participants in the menopausal hormone therapy (MHT) group compared with those in the risedronate (RISE) group. p -Values were calculated using the log-rank test. CI, confidence interval; HR, hazard ratio. * $p < 0.01$ vs. baseline. Used with the permission of Yoon [20].

death via an antioxidant mechanism by downregulating both the extrinsic and intrinsic apoptosis pathways [24].

According to the WHI study, MPA adversely affects CEE's effects on CHD [8]. As for stroke, both CEE alone and in combination with MPA increased the risk of ischemic but not hemorrhagic stroke [9,10]. A French observational study reported that the effects of MHT on the risk of ischemic stroke might differ according to the estrogen dose and route of administration and also the type of progestogen; MHT using low-dose oral estrogen, transdermal estrogen or natural progesterone did not increase the risk likely due to less stimulation of blood coagulation [25].

Even though both share major risk factors, the impacts of oral MHT on CHD and stroke are quite different in recently postmenopausal women [12]. Dyslipidemia is more dominant in CHD, while hypertension is more dominant in stroke. The responses of risk factors to estrogen depend on the route of administration and are mainly mediated by the hepatic first-pass effect. Parenteral (but not oral) estrogen is reported to reduce blood pressure [26].

We conducted a placebo-controlled RCT to investigate the effects of oral MP4 (100 mg/day) combined with PEG (0.1% 1.0 g/day) for 12 weeks on hemodynamics in postmenopausal women ($n=52$, mean age 57 years) with grade 1 hypertension [26]. In this study, PEG had no effect on 24-h ambulatory blood pressure, and MP4 did not influence PEG's effects. We also studied the impacts of MHT on systemic arterial stiffness, which was evaluated using brachial-ankle pulse-wave velocity. After controlling for the baseline value, the linear regression analysis indicated that PEG improved arterial stiffness by significantly decreasing the pulse-wave velocity, and MP4 exerted a neutral effect on PEG. These findings might provide an additional rationale for the study to test the effect of parenteral estrogen and MP4 on stroke risk in early postmenopausal women.

Alzheimer's disease

Dementia refers to multiple cognitive deficits that are progressive and severe enough to cause impairment in activities of daily living (ADLs). Alzheimer's disease (AD) is a major type of dementia. Mild cognitive impairment is an intermediate stage between normal aging and dementia, and is characterized by the preservation of ADLs despite objective impairments in cognition. Unfortunately, no pharmacologic treatment to date has been proven clinically effective in both conditions.

CEE made no difference in cognition or ADLs in hysterectomized women with mild to moderate AD [27]. Interestingly, transdermal E2 therapy at a higher dosage (100 µg/day) was reported to have a positive influence on cognition [28]. Progestogen also shows widely different effects by preparation; MPA caused detrimental effects on cognition [11,29], whereas MP4 might be beneficial [30].

Our group conducted an open-label RCT to explore the therapeutic potential of CEE and MP4 therapy for 6 months in postmenopausal women ($n=55$, mean age 70 years) with mild to moderate AD, as compared with tacrine, a cholinesterase

inhibitor [31]. There was no difference in cognition between the two groups. A *t*-test showed that ADLs significantly worsened in the tacrine group, but ADLs remained unchanged in the MHT group. Importantly, ADLs differed significantly between the two groups in time trends (Figure 2(A)) and changes from baseline at 6 months (Figure 2(B)) as analyzed by repeated-measures analysis of variance and *t*-test, respectively.

Another placebo-controlled RCT was performed to test the therapeutic efficacy of PEG (0.1% 2 mg/day) plus oral MP4 therapy for 24 months in postmenopausal women ($n=37$, mean age 70 years) with multiple-domain, amnesic-subtype mild cognitive impairment [32]. Analyzed using a generalized estimating equation, the Montreal Cognitive Assessment (MoCA) test, a sensitive tool for assessing global cognition in mild cognitive impairment, revealed a significant deterioration in the placebo group but not in the MHT group (Figure 2(C)). There was a significant difference in the time trend between the two groups after adjusting for the $\epsilon 4$ allele. In addition, MoCA scores in the MHT groups were significantly better at 18 and 24 months than those in the placebo group.

All-cause mortality

All-cause mortality (ACM) is one definite endpoint to use to assess a complex balance of benefits and risks. We updated the ACM data in postmenopausal women younger than 60 years by performing an adaptive meta-analysis [33] of four MHT trials [34]. Two trials [35,36] were conducted in healthy women, and the other two [37,38] were carried out in women with a specific diagnosis. We also used cumulative data, when available, including intervention and post-intervention follow-up [35,36,38], which could provide more robust evidence.

During the intervention phase, ACM was significantly decreased with MHT by 41%. Notably, mortality seemed to be reduced in younger menopausal women with chronic disease [37] and CHD [38]. An analysis of the cumulative data revealed that estrogen therapy decreased the ACM significantly by 20%, but estrogen-progestogen therapy did not reach statistical significance. Our results are consistent with a recent review on the impact of MHT on overall mortality [39].

Discussion and conclusion

E2 levels fluctuate dramatically in reproductive-age women within the normal menstrual cycle between 20 and 400 pg/ml. Therapeutic levels of E2 for the prevention of late problems of OHD have not yet been defined. Further study is needed to refine the optimal MHT regimen in terms of dose, route of administration and preparation, which might benefit even older postmenopausal women considering MHT.

A primary prevention trial for AD is not practically feasible due to the large sample size and long duration of follow-up required. Of note, transdermal E2 and oral MP4 therapy initiated shortly after menopause might reduce amyloid- β deposition in the brain in cognitively normal women [40]. The WHI study also reported a decline in dementia mortality with CEE therapy [36]. These findings in conjunction with the timing

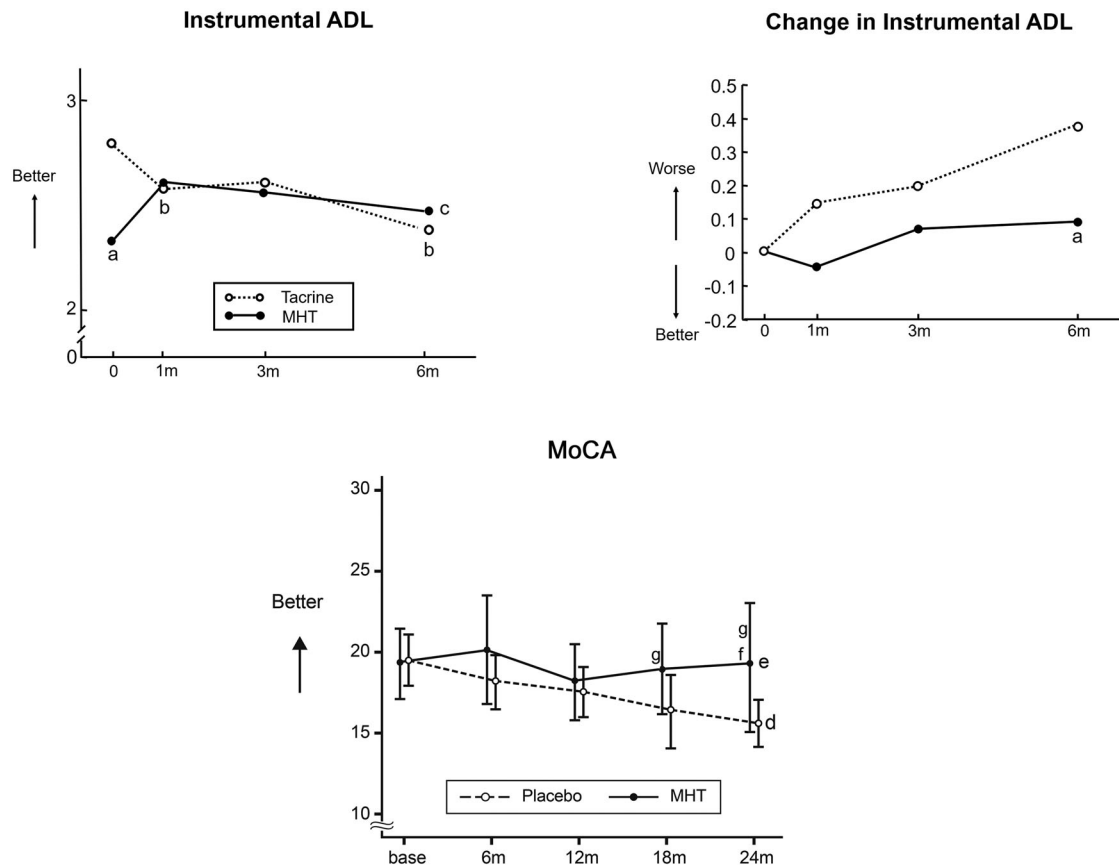


Figure 2. Impacts of menopausal hormone therapy (MHT) on activities of daily living (ADLs) in women with Alzheimer's disease and general cognition in women with mild cognitive impairment. Mean changes in the instrumental ADL (A), changes from baseline on the instrumental ADL (B) and mean changes in the Montreal Cognitive Assessment (MoCA) (C). ^a $p < 0.05$ vs. tacrine. ^b $p < 0.05$ vs. baseline within groups. ^cSignificant difference in time trend compared to tacrine. ^dSignificant time trend within groups. ^eSignificant difference in time trend compared to placebo, after controlling for the $\epsilon 4$ allele. ^f $p < 0.05$ vs. placebo. ^g $p < 0.05$ vs. placebo after adjusting for the $\epsilon 4$ allele. Vertical bars indicate 95% confidence intervals. Used with the permission of Yoon [31,32].

hypothesis strongly suggest MHT as a possible strategy for AD prevention in recently postmenopausal women.

Based on 20 years of follow-up, the WHI demonstrated that CEE decreased the risk of breast cancer, one of the major barriers against MHT, in terms of both incidence and mortality [41]. At the same time, CEE plus MPA increased the breast cancer incidence but not the breast cancer mortality [41]; however, estrogen-progestogen therapy using other progestogens, such as norethindrone acetate [35] and MP4 [42], had no influence on breast cancer incidence.

MHT improves menopause-related quality of life [43] and decreases ACM in younger postmenopausal women [34,39]. It may be time to cautiously reconsider MHT in early postmenopausal women for primary prevention of the late problems of OHD [44–46].

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